train, the disparity is small—Martin claims that Scratchy got the light less than a trillionth of a second after Itchy—but were the train moving faster, near light speed, the time difference would be substantial.

Think about what this means for the flip-book pages slicing up a region of spacetime. Since observers moving relative to each other do not agree on what things happen simultaneously, the way each of them will slice a block of spacetime into pages—with each page containing all events that happen at a given moment from each observer's perspective — will not agree, either. Instead, observers moving relative to each other cut a block of spacetime up into pages, into time slices, in different but equally valid ways. What Lisa and Marge found for space, Einstein found for spacetime.

Angling the Slices

The analogy between street/avenue grids and time slicings can be taken even further. Just as Marge's and Lisa's designs differed by a rotation, Apu's and Martin's time slicings, their flip-book pages, also differ by a rotation, but one that involves both space and time. This is illustrated in Figures 3.4a and 3.4b, in which we see that Martin's slices are rotated relative to Apu's, leading him to conclude that the duel was unfair. A critical difference of detail, though, is that whereas the rotation angle between Marge's and Lisa's schemes was merely a design choice, the rotation angle between Apu's and Martin's slicings is determined by their relative speed. With minimal effort, we can see why.

Imagine that Itchy and Scratchy have reconciled. Instead of trying to shoot each other, they just want to ensure that clocks on the front and back of the train are perfectly synchronized. Since they are still equidistant from the gunpowder, they come up with the following plan. They agree to set their clocks to noon just as they see the light from the flaring gunpowder. From their perspective, the light has to travel the same distance to reach either of them, and since light's speed is constant, it will reach them simultaneously. But, by the same reasoning as before, Martin and anyone else viewing from the platform will say that Itchy is heading toward the emitted light while Scratchy is moving away from it, and so Itchy will receive the light signal a little before Scratchy does.